

JEE MAIN REVISION MASTER: S-BLOCK ELEMENTS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet S Block Elements - Core Application	8 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet S Block Elements - Core Application (High Yield — 8 PYQs)

Core Principle: Key problem pattern in Kcet S Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following nitrate will decompose to give NO_2 on heating ?..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "The property of the alkaline earth metals that increases with their atomic number is..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "The oxide of potassium that does not exist is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which one of the following nitrate will decompose to give NO_2 on heating ?

- (A) NaNO_3
- (B) KNO_3
- (C) RbNO_3
- (D) LiNO_3

Correct Answer: None

Solution Analysis: Among the given nitrates, only LiNO_3 will decompose to give NO_2 when heated. In contrast, the nitrates of sodium (Na), potassium (K), and rubidium (Rb) decompose to release oxygen instead. The decomposition reactions are as follows: For lithium nitrate (LiNO_3): $2\text{LiNO}_3 \xrightarrow{\Delta} 2\text{Li}_2\text{O} + 4\text{NO}_2 + \text{O}_2$ For sodium, potassium, and rubidium nitrates (MNO_3), where $\text{M}=\text{Na}, \text{K}, \text{Rb}$): $2\text{MNO}_3 \xrightarrow{\Delta} 2\text{MNO}_2 + \text{O}_2$ Here, LiNO_3 uniquely produces NO_2 along with the metal oxide and oxygen.